Semester 1 Examinations 2011

Exam Code(s)	1MF1, 1SD1, 2CS1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), M.Sc in Software Design & Development, BSc (Computing Studies)
Module Code(s)	CT853, CT232
Module(s)	Algorithmics & Logical Methods, Methodology
Paper No.	
Repeat Paper	
External Examiner(s)	Prof. Michael O'Boyle
Internal Examiner(s)	Prof. Gerard Lyons Dr. Jim Duggan *Martina Fox

Instructions: Answer any 3 questions. All questions carry equal marks

Duration	2 hours
No. of Pages	2
Discipline(s)	Information Technology
Course Co-ordinator(s)	

Requirements:
MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Release to Library: Yes ☐ No ☐

Q1 a Given an array of size 7 with integer data containing: 40, 2, 7, 13, 5, 20, 1 write down, step by step, the position of each number as sorting is carried out using the following sorting algorithms:

1. Selection Sort
2. Bubble sort' [10]

b. Which method runs fastest for an array with all keys identical, selection sort or insertion sort? [5]

c Write the pseudocode which allows you to search a list [L1,L2…LN] for an element X. Given that each element is unique, successively compare X to each element in the list L1,, L2..LN until X is found or the end of the list is reached. [10]

Q2 a What is an algorithm and why do we study them? [5]

b What are the desirable properties of an algorithm [5]

c Write a algorithm that, given an array a[] of *N* distinct integers, finds a *local maximum*: an index i such that a[i] < a[i-1] and a[i] < a[i+1]. Your program should use ~ $lg_2 N$ comparisons in the worst case [15]

Q3 a Explain the Divide & Conquer approach to algorithm design. Give an example (using pseudocode) of a Divide & Conquer algorithm to calculate X^n , where n is a positive integer [10]

b Using Karatsuba's algorithm multiply 1234 by 5678 .Describe the algorithm explain why it is more efficient than 'standard' multiplication and give (in 'big oh' notation) its complexity

[15]

Q4 A container truck has total (weight) capacity of 7 kilo Tonnes. In a port there are five loads it can collect , each with a different weight and value:

Load	**A**	**B**	**C**	**D**	**E**
Weight (Kilo Tonnes)	2	5	1	1	3
Value (Millions of Euro)	11	24	7	8	20

The objective is to decide which loads to take to maximise the *Value.* Using *Dynamic Programming,* calculate which loads should be chosen. You must show your calculations in detail by writing the table of values calculated by the algorithm

[25]